WORKSHEET 7.4

Reasoning from a Slope Field

Unit 7: Differential Equations

Key ideas

- An **equilibrium solution** is a constant solution: a horizontal line in the slope field.
- A solution curve cannot cross an equilibrium. The sign of $\frac{dy}{dt}$ on each side tells you whether solutions approach it or move away.
- $\frac{d^2y}{dx^2}$ is found by differentiating the equation and substituting for $\frac{dy}{dx}$. Its sign gives concavity.
- For a logistic-type equation, growth is fastest halfway between the equilibria, and the solution has a point of inflection there.

Worked examples

Example 1 · One equilibrium

Describe the long-run behaviour of solutions of $\frac{dy}{dt} = 3 - y$.

Solution. Equilibrium $y = 3$. Above it the derivative is negative; below it, positive.

Every solution approaches 3.

Example 2 · Two equilibria

For $\frac{dy}{dt} = y(5 - y)$, describe the solutions with $y(0) = 1$ and with $y(0) = 7$.

Solution. Equilibria 0 and 5. Between them the derivative is positive; above 5 it is negative.

The first increases toward 5; the second decreases toward 5.

Example 3 · A second derivative

For $\frac{dy}{dx} = 3x - y$, find $\frac{d^2y}{dx^2}$ in terms of x and y.

Solution. Differentiate: $3 - \frac{dy}{dx}$, then substitute.

$3 - 3x + y$.

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Practice

CORE

1. Find the equilibrium solutions of $\frac{dy}{dt} = y^2 - 16$.

2. A solution of $\frac{dy}{dt} = 0.4(20 - y)$ has $y(0) = 5$. Find $\lim_{t \to \infty} y(t)$.

3. For the same equation, describe the solution with $y(0) = 30$.

4. For $\frac{dy}{dx} = x^2 + y$, find $\frac{d^2y}{dx^2}$ in terms of x and y.

EXAM LEVEL

5. For $\frac{dy}{dt} = \frac{y(10 - y)}{5}$ with $0 < y < 10$, at what value of y is the solution growing fastest, and how fast?

6. For the equation in Question 5, at what value of y does a solution have a point of inflection? Give a reason.

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7. A solution of $\dfrac{dy}{dx} = x - y^2$ passes through $(4, 2)$. Show that it has a relative minimum there.

8. For $\dfrac{dy}{dx} = 3x - y$, describe the region of the plane in which solution curves are concave up.

Question 9 · Free-response practice

A fish population P, in thousands, satisfies $\dfrac{dP}{dt} = 0.2P\left(1 - \dfrac{P}{50}\right)$, where t is in years.

(a) Find the equilibrium solutions.

(b) If $P(0) = 10$, find $\lim\limits_{t \to \infty} P(t)$ and explain why P is increasing for all t.

(c) Find the population at which the fish are increasing fastest, and the rate of increase then.

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Worked solutions

1. $y = 4$ and $y = -4$.
2. 20.
3. It decreases, approaching 20 from above.
4. $2x + \dfrac{dy}{dx} = 2x + x^2 + y$.
5. At $y = 5$, halfway between the equilibria; the rate is $\dfrac{5(5)}{5} = 5$.
6. At $y = 5$: $\dfrac{d^2y}{dt^2} = \dfrac{10 - 2y}{5} \cdot \dfrac{dy}{dt}$ changes sign there.
7. $\dfrac{dy}{dx} = 4 - 4 = 0$, and $\dfrac{d^2y}{dx^2} = 1 - 2y\dfrac{dy}{dx} = 1 > 0$.
8. $\dfrac{d^2y}{dx^2} = 3 - 3x + y > 0$: above the line $y = 3x - 3$.
9. (a) $P = 0$ and $P = 50$.
   (b) 50. For $0 < P < 50$ both factors are positive, so $\dfrac{dP}{dt} > 0$, and the solution cannot cross the equilibrium $P = 50$.
   (c) At $P = 25$ thousand: $0.2(25)(0.5) = 2.5$ thousand fish per year.

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